

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5132

5 / 5 / 3 / 3 / 3 Credits: 4

Program Core

Source-grounded

Operating System

□ Recognize the basic operating system concepts.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5132 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5132.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Engineering / Computer Hardware Engineering /
Course code	5132
Course title	Operating System
Semester	5 / 5 / 3 / 3 / 3 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

20 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ Recognize the basic operating system concepts.
- □ Provide a comprehensive introduction to process management, memory management, I/O management,

Course outcomes

CO	Official outcome	Study level
CO1	10 Understanding functions.	See official syllabus
CO2	Explain process and process management. 16 Understanding	See official syllabus
CO3	Illustrate memory management schemes. 20 Understanding Explain file organization and disk scheduling	See official syllabus
CO4	12 Understanding algorithms	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the significance of operating systems and its functions	4	As specified
Module 2	Describe process & process management	6	As specified
Module 3	Illustrate memory management schemes.	5	As specified
Module 4	Explain file organization and disk scheduling algorithms	5	As specified

MODULE 1

Outline the significance of operating systems and its functions

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the significance of operating systems and its functions
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding. application software. Explain System software-assembler, compiler,

1 Understanding. application software. Explain System software-assembler, compiler, is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. application software. Explain System software-assembler, compiler, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. application software. explain system software-assembler, compiler, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding. application software. Explain System software-assembler, compiler, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding. application software. Explain System software-assembler, compiler, definition/meaning definition/meaning . steps, application application . Technical terms English- .

▼ interpreter and operating system with 3 Understanding. examples

interpreter and operating system with 3 Understanding. examples is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify interpreter and operating system with 3 Understanding. examples using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpreter and operating system with 3 understanding. examples should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What interpreter and operating system with 3 Understanding. examples means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇന്റർപ്രീറ്റർ and operating system with 3 Understanding. examples ഇന്റർപ്രീറ്റർ definition/meaning ഇന്റർപ്രീറ്റർ. ഇന്റർപ്രീറ്റർ ഇന്റർപ്രീറ്റർ, steps, application ഇന്റർപ്രീറ്റർ ഇന്റർപ്രീറ്റർ. Technical terms English-ഇന്റർപ്രീറ്റർ ഇന്റർപ്രീറ്റർ.

▼ Explain the functions of Operating System 3 Understanding.

Explain the functions of Operating System 3 Understanding. is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the functions of Operating System 3 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the functions of operating system 3 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain the functions of Operating System 3 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the functions of Operating System 3 Understanding. definition/meaning. steps, application. Technical terms English-.

▼ Summarise different types of operating system 3 Understanding.

Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.

Summarise different types of operating system 3 Understanding.

Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. is an official topic in Module 1 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise different types of operating system 3 understanding. introduction to software: system software & application software, examples-system software - assembler, loader, compilers, interpreters and their functions, comparison. operating system - definition - functions of operating system - different types of operating systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
 Summarise different types of operating system
 3 Understanding. Introduction to software: System software & application software, examples-System software – assembler, loader, compilers, interpreters and their functions, comparison. Operating system – definition – functions of Operating system – different types of Operating Systems – batch systems – multiprogramming systems – time sharing – multiprocessor systems – real time systems.
 definition/meaning
 , steps, application
 . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Describe process & process management

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe process & process management
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding structure Explain various states of a process with state

2 Understanding structure Explain various states of a process with state is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding structure Explain various states of a process with state using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding structure explain various states of a process with state should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding structure Explain various states of a process with state means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding structure Explain various states of a process with state definition/meaning . . . , steps, application Technical terms English-

▼ 2 Understanding diagram Classify CPU schedulers and Illustrate

2 Understanding diagram Classify CPU schedulers and Illustrate is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding diagram Classify CPU schedulers and Illustrate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding diagram classify cpu schedulers and illustrate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding diagram Classify CPU schedulers and Illustrate means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding diagram Classify CPU schedulers and Illustrate definition/meaning. 2026 2026 2026, steps, application 2026 2026. Technical terms English- 2026 2026.

▼ 6 Understanding scheduling algorithms

6 Understanding scheduling algorithms is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding scheduling algorithms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding scheduling algorithms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding scheduling algorithms means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 6 Understanding scheduling algorithms
 പഠിക്കേണ്ടതെന്തെന്ന് definition/meaning പഠിക്കേണ്ടതെന്തെന്ന്. പഠിക്കേണ്ട പഠിക്കേണ്ട
 പഠിക്കേണ്ട, steps, application പഠിക്കേണ്ട പഠിക്കേണ്ട പഠിക്കേണ്ട. Technical terms
 English- പഠിക്കേണ്ട പഠിക്കേണ്ട പഠിക്കേണ്ട.

▼ Show resource allocation graph

Show resource allocation graph is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Show resource allocation graph using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, show resource allocation graph should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Show resource allocation graph means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
 Show resource allocation graph
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ Demonstrate deadlock and its causes

Demonstrate deadlock and its causes is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate deadlock and its causes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate deadlock and its causes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate deadlock and its causes means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-☐☐ Demonstrate deadlock and its causes ☐☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐ ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

▼ **Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes- Process synchronization and critical section management.**

Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes-Process synchronization and critical section management. is an official topic in Module 2 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various

scheduling algorithms-Resource allocation graph-deadlock-its causes-
Process synchronization and critical section management. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain process synchronization 2 understanding define process - process control block (pcb) and its general structure - different states of a process with the help of state diagram.- schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- various scheduling criteria - fcfs, sjf, priority, and rr scheduling algorithms with gantt charts -comparison of various scheduling algorithms-resource allocation graph-deadlock-its causes-process synchronization and critical section management. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes- Process synchronization and critical section management. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain Process synchronization 2 Understanding Define process - process control block (PCB) and its general structure - different states of a process with the help of state diagram.- Schedulers - long, medium and short term- difference between preemptive and non-preemptive scheduling- Various scheduling criteria - FCFS, SJF, Priority, and RR scheduling algorithms with Gantt charts -comparison of various scheduling algorithms-Resource allocation graph-deadlock-its causes-Process synchronization and critical section management.

പ്രക്രിയയുടെ definition/meaning. പ്രക്രിയയുടെ states, steps, application. Technical terms English-ൽ എഴുതുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Illustrate memory management schemes.

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate memory management schemes.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Define memory management.

Define memory management. is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define memory management. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define memory management. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define memory management. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Define memory management.

മെമ്മറി മാനേജ്മെന്റ് definition/meaning. മെമ്മറി മാനേജ്മെന്റ്, steps, application. Technical terms English-ൽ മെമ്മറി മാനേജ്മെന്റ്.

▼ Show various address binding schemes

Show various address binding schemes is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Show various address binding schemes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, show various address binding schemes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Show various address binding schemes means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Show various address binding schemes
 ഓരോന്നിനും definition/meaning നൽകുക. ഓരോന്നിനും
 ഓരോന്നിനും, steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ Explain memory management techniques

Explain memory management techniques is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain memory management techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain memory management techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain memory management techniques means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain memory management techniques definition/meaning. , steps, application . Technical terms English- .

▼ Illustrate memory allocation strategies.

Illustrate memory allocation strategies. is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate memory allocation strategies. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate memory allocation strategies. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate memory allocation strategies. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓസറ്റംസ് ഇല്ലാസ്റ്റേറ്റ് മെമറി അലോക്കേഷൻ സ്ട്രാറ്റജികൾ. ഓസറ്റംസ് ഓസറ്റംസ് ഓസറ്റംസ്, steps, application ഓസറ്റംസ് ഓസറ്റംസ് ഓസറ്റംസ്. Technical terms English-ഓ ഓസറ്റംസ് ഓസറ്റംസ് ഓസറ്റംസ്.

▼ **Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings - compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation - fixed partition and variable partition - Allocation Strategies - first fit, best fit and worst fit - Define fragmentation - internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing.**

Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings – compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation – fixed partition and variable partition – Allocation Strategies - first fit, best fit and worst fit - Define fragmentation – internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing. is an official topic in Module 3 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings – compile,

link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation – fixed partition and variable partition – Allocation Strategies - first fit, best fit and worst fit - Define fragmentation – internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU - Thrashing. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate page replacement algorithms. 6 understanding memory management - different address bindings – compile, link and run time bindings. - difference between logical address and physical address - contiguous memory allocation – fixed partition and variable partition – allocation strategies - first fit, best fit and worst fit - define fragmentation – internal and external, and solutions - paging and paging hardware - segmentation, advantages of segmentation over paging- concept of virtual memory - demand paging - page-faults and how to handle page faults. - page replacement algorithms: fifo, optimal, lru -thrashing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings – compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation – fixed partition and variable partition – Allocation Strategies - first fit, best fit and worst fit - Define fragmentation – internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging-

Aspect	What to prepare
	Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU - Thrashing. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Demonstrate page replacement algorithms. 6 Understanding Memory management - Different address bindings – compile, link and run time bindings. - Difference between logical address and physical address - Contiguous memory allocation – fixed partition and variable partition – Allocation Strategies - first fit, best fit and worst fit - Define fragmentation – internal and external, and solutions - Paging and paging hardware - Segmentation, advantages of segmentation over paging- Concept of virtual memory - Demand paging - Page-faults and how to handle page faults. - Page replacement algorithms: FIFO, optimal, LRU -Thrashing. ഫസ്റ്റ് ഫിറ്റ്/മിസ്റ്റ് ഫിറ്റ്/വേസ്റ്റ് ഫിറ്റ് definition/meaning. ഫസ്റ്റ് ഫിറ്റ്, മിസ്റ്റ് ഫിറ്റ്, വേസ്റ്റ് ഫിറ്റ്, steps, application. Technical terms English- മിസ്റ്റ് ഫിറ്റ്, വേസ്റ്റ് ഫിറ്റ്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain file organization and disk scheduling algorithms

This module is presented from the official syllabus scope for Operating System. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain file organization and disk scheduling algorithms
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 2 Remembering operations

2 Remembering operations is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering operations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering operations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering operations means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Remembering operations

പ്രതിരോധനം പ്രതിരോധനം definition/meaning പ്രതിരോധനം. പ്രതിരോധനം പ്രതിരോധനം പ്രതിരോധനം, steps, application പ്രതിരോധനം പ്രതിരോധനം പ്രതിരോധനം. Technical terms English- പ്രതിരോധനം പ്രതിരോധനം.

▼ Explain different file organizations

Explain different file organizations is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different file organizations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different file organizations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different file organizations means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 Explain different file organizations
 definition/meaning.
 steps, application
 Technical terms
 English-
 .

▼ Classify directory structures

Classify directory structures is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify directory structures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify directory structures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify directory structures means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-☐☐ Classify directory structures
☐☐☐☐☐☐☐☐☐☐ ☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐ ☐☐☐☐☐
☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐ ☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms
English-☐ ☐☐☐☐☐☐☐☐☐☐☐☐.

▼ Illustrate file allocation methods

Illustrate file allocation methods is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate file allocation methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate file allocation methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate file allocation methods means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓസറേറ്റിംഗ് സിസ്റ്റം ഇല്ലാത്ത ഫയൽ അലോക്കേഷൻ രീതികൾ
 ഓസറേറ്റിംഗ് സിസ്റ്റം definition/meaning ഓസറേറ്റിംഗ് സിസ്റ്റം. ഓസറേറ്റിംഗ് സിസ്റ്റം
 ഓസറേറ്റിംഗ്, steps, application ഓസറേറ്റിംഗ് ഓസറേറ്റിംഗ് ഓസറേറ്റിംഗ്. Technical terms
 English-ഓ ഓസറേറ്റിംഗ് ഓസറേറ്റിംഗ്.

▼ **Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm**

Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm is an official topic in Module 4 of Operating System. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts – sequential and indexed. Different directory structures – single level, two-level, and tree structured directories. - Different allocation methods – contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain disk scheduling algorithms. 3 understanding file system - concept of file and directory - various file operations - file organization concepts – sequential and indexed. different directory structures – single level, two-level, and tree structured directories. - different allocation methods – contiguous, linked and indexed allocations. various disk scheduling algorithms-fcfs, sstf, scan, c-scan, look & c-look. text / reference t/r book title/author

abraham silberschatz, peter gaer galvin and greg gagne, operating system t1 concepts, wiley publications-eighth edition william stallings, operating systems-internals and design principles, r1 pearson publications- seventh edition r2 andrew s tanenbaum, herbert bos, modern operating systems, pearson, 4/e, online resources sl.no website link 1

<https://nptel.ac.in/courses/106/105/106105214/#> 2

https://www.tutorialspoint.com/operating_system/index.htm should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain disk scheduling algorithms. 3 Understanding File system - Concept of file and directory - Various file operations - File organization concepts - sequential and indexed. Different directory structures - single level, two-level, and tree structured directories. - Different allocation methods - contiguous, linked and indexed allocations. Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look. Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-Eighth Edition William Stallings, Operating Systems-Internals and Design Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105214/# 2 https://www.tutorialspoint.com/operating_system/index.htm means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic

with a related but different term.

Malayalam support: Module 4- Explain disk scheduling algorithms. 3
 Understanding File system - Concept of file and directory - Various file
 operations - File organization concepts – sequential and indexed. Different
 directory structures – single level, two-level, and tree structured directories.
 - Different allocation methods – contiguous, linked and indexed allocations.
 Various disk scheduling algorithms-FCFS, SSTF, Scan, C-Scan, Look & C-Look.
 Text / Reference T/R Book Title/Author Abraham Silberschatz, Peter Gaer
 Galvin and Greg Gagne, Operating System T1 Concepts, Wiley Publications-
 Eighth Edition William Stallings, Operating Systems-Internals and Design
 Principles, R1 PEARSON Publications- Seventh Edition R2 Andrew S
 Tanenbaum, Herbert Bos, Modern Operating Systems, Pearson, 4/e, Online
 Resources Sl.No Website Link 1
<https://nptel.ac.in/courses/106/105/106105214/#> 2
https://www.tutorialspoint.com/operating_system/index.htm
 definition/meaning . , steps,
 application . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 1 Understanding. application software. Explain System software-assembler, compiler,. (3 marks)**

► **Define or explain interpreter and operating system with 3 Understanding. examples. (3 marks)**

► **Define or explain Explain the functions of Operating System 3 Understanding.. (3 marks)**

► **Define or explain Summarise different types of operating system 3 Understanding. Introduction to software: System software & application software, examples-System software - assembler, loader, compilers, interpreters and their functions, comparison. Operating system - definition - functions of Operating system - different types of Operating Systems - batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.. (3 marks)**

Module 2

► **Define or explain 2 Understanding structure Explain various states of a process with state. (3 marks)**

► **Define or explain 2 Understanding diagram Classify CPU schedulers and Illustrate. (3 marks)**

► **Define or explain 6 Understanding scheduling algorithms. (3 marks)**

► **Define or explain Show resource allocation graph. (3 marks)**

Module 3

► **Define or explain Define memory management.. (3 marks)**

► **Define or explain Show various address binding schemes. (3 marks)**

► **Define or explain Explain memory management techniques. (3 marks)**

► **Define or explain Illustrate memory allocation strategies.. (3 marks)**

Module 4

► **Define or explain 2 Remembering operations. (3 marks)**

► **Define or explain Explain different file organizations. (3 marks)**

► **Define or explain Classify directory structures. (3 marks)**

► **Define or explain Illustrate file allocation methods. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding. application software. Explain System software-assembler, compiler,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding structure Explain various states of a process with state.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Define memory management..**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **2 Remembering operations.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

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QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/106/105/106105214/#>
- https://www.tutorialspoint.com/operating_system/index.htm

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5132. Verify institutional updates against the current official curriculum.